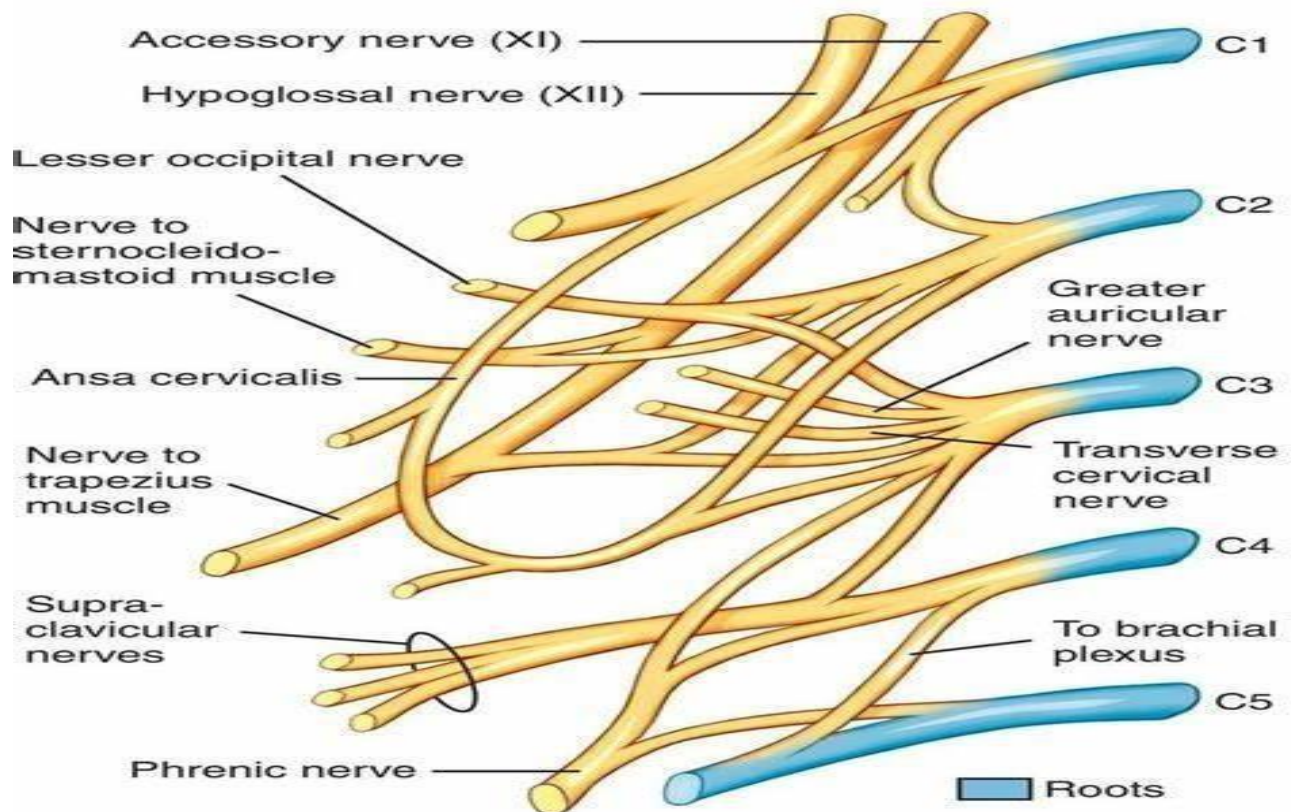




Lecture (5): Blood supply & lymphatic drainage of the head & neck

□ Arteries:

Common carotid

artery:

▪ Begin:

The **left common carotid artery** arises from the arch of the aorta. The **right common carotid artery** arises from brachiocephalic artery.

- **Course:** It ascends inside the carotid sheath **lateral** to the trachea and esophagus.
- **End:** at the upper border of thyroid cartilage opposite **C4**, where it splits into its two **terminal branches: the external and internal carotid arteries.**
- The **carotid body** and the **carotid sinus** are two specialized sensory organs associated with the bifurcation of the common carotid artery and its internal branch.

External carotid artery (ECA):

- **Begin:** It originates as one of the two terminal branches of the common carotid artery at the upper border of thyroid cartilage opposite C4 vertebra.
- **Course:** It ascends outside the carotid sheath in the carotid triangle of the neck. Then ascends inside the parotid gland.
- **End:** by giving two terminal branches; **superficial temporal & maxillary arteries** behind the neck of the mandible.

- **Branches:**

- **3 anterior branches:**

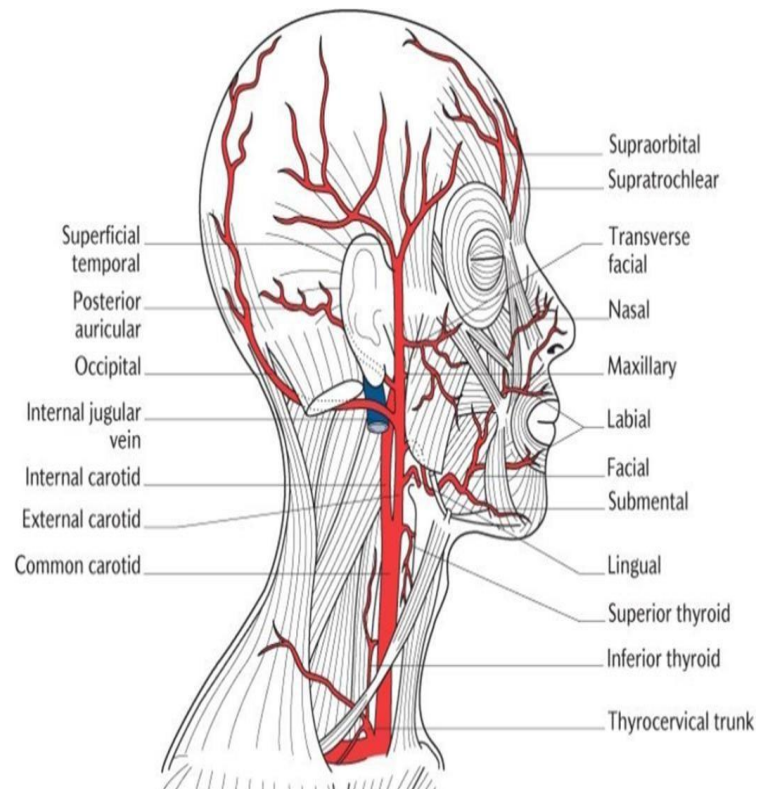
- 1) **Superior thyroid artery** arises lateral to thyroid cartilage supplies the thyroid gland.
- 2) **Lingual artery** gives tonsillar, dorsal lingual, deep lingual and sublingual arteries.
- 3) **Facial artery** gives rise to ascending palatine, submental, glandular, and tonsillar branches.

- **Two posterior branches:**

- 1- **Occipital artery** arises inferiorly to the posterior belly of the digastric muscle.
- 2- **Posterior auricular artery** arises superiorly to the posterior belly of the digastric muscle.

- **3 ascending branches:**

- 1- **Maxillary artery** is one of the two terminal branches of the external carotid artery in the parotid gland.
- 2- **Superficial temporal artery** is the other terminal branch of external carotid artery in the parotid gland.



3- **Ascending pharyngeal artery** arises on the medial surface of external carotid artery near its origin.

Maxillary artery:

- **Origin:** It arises as one of two terminal branches of the ECA behind the neck of mandible.
- **Course:** It is divided into three parts by the lateral pterygoid muscle:
 - **1st part:** artery, which is medial to mandibular neck.
 - **2nd part:** deep to the muscle.
 - **3rd part:** between the two heads of the muscle.
- **Branches:**

➤ **From the 1st part:**

- 1- **Deep auricular artery** supplies the external auditory canal and external tympanic membrane.
- 2- **Anterior tympanic artery** forms a vascular circle around the tympanic membrane with the stylomastoid branch of the posterior auricular artery.
- 3- **Accessory meningeal artery** supplies the dura.
- 4- **Middle meningeal artery** supplies the dura.
- 5- **Inferior alveolar artery** supplies the lower jaw.

➤ **From the 2nd part:** 5 muscular branches to the muscles of mastication.

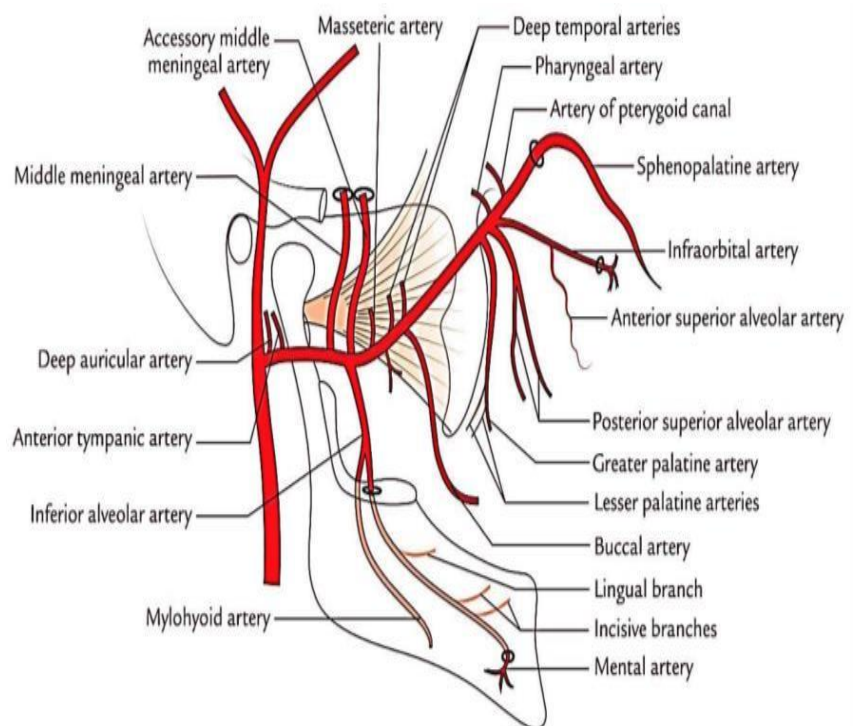
➤ **From the 3rd part:**

1- **Greater palatine artery** supplies the hard palate.

2- **Sphenopalatine artery** supplies the nasal cavities.

3- **Posterior superior alveolar artery** supplies upper molar & premolar teeth.

4- **The artery to pterygoid canal** supplies the roof of pharynx.



- 5- **Infra-orbital artery** supplies canine & incisors of the upper jaw via the anterior superior alveolar artery and ends by supplying the face.

Subclavian artery:

- **Origin:**

It arises from the **arch of aorta** on the left side & **brachiocephalic artery** on the right side

- **Course:**

It has an arched course at the root of the neck.

The **Scalenus anterior muscle** divides the artery into **3 parts**:

1st part is medial to the muscle; **2nd part** is deep to the muscle & **3rd part** is lateral to the muscle.

- **End:** at the outer border of the 1st rib by becoming an **axillary artery**.

- **Branches:**

1- From the **1st part**:

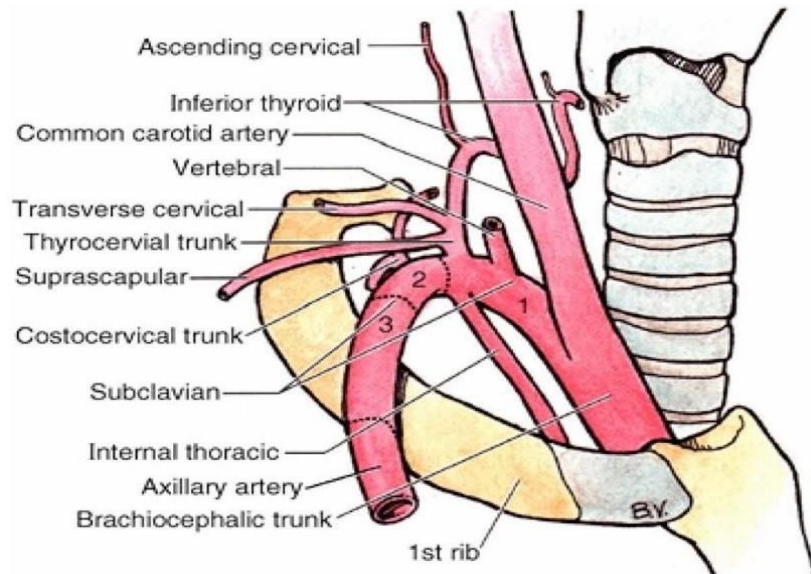
- **Internal thoracic (mammary) artery**.
- **Vertebral artery**.

- **Thyrocervical trunk**: it gives off: **Suprascapular artery**, **Transverse cervical artery** and **Inferior thyroid artery**.

2- From the **2nd part**: Only one branch.

- **Costo-cervical trunk** which gives off: **Superior intercostal artery** & **deep cervical artery**.

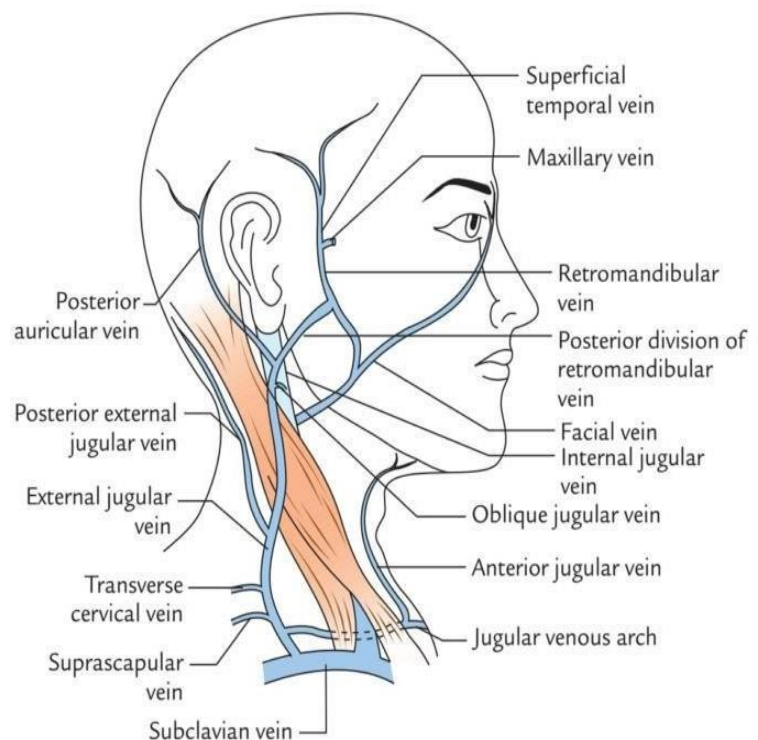
3- From the **3rd part**: **no branches**.



Veins of the neck:

1- External jugular vein:

- It is formed by union of the **posterior division** of the **retromandibular vein** and the **posterior auricular vein** just inferior to the parotid gland.
- It descends on the lateral surface of sternomastoid muscle.
- In the posterior triangle, it empties into the **subclavian vein**.
- **Its tributaries:** anterior jugular, Suprascapular & transverse cervical veins



Anterior jugular vein:

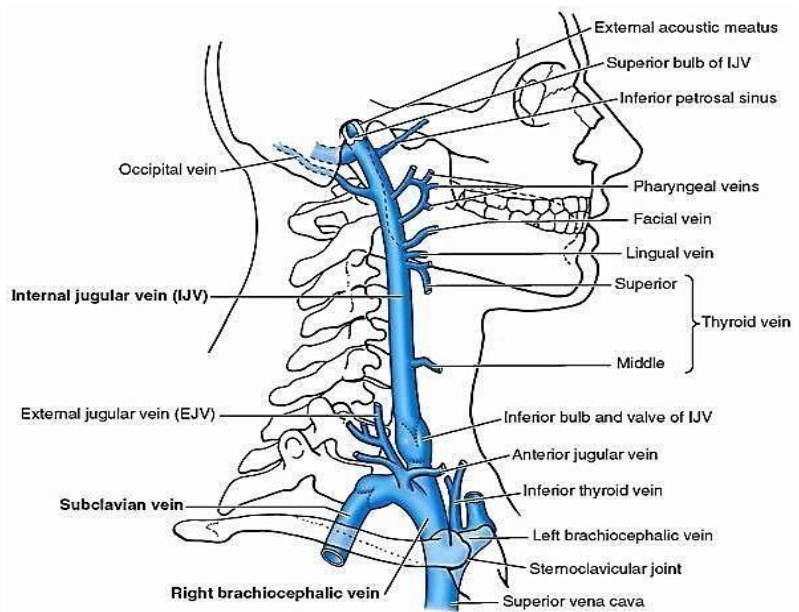
- **Begin** in the submental triangle by union of submental veins. They descend near the midline to drain laterally into **external jugular vein**.
- The two anterior jugular veins are connected above the jugular notch by the jugular venous arch in the suprasternal space.

3- Internal jugular vein:

- It begins as a downward continuation of the **sigmoid sinus** through the **jugular foramen**.
- It descends inside the **carotid sheath** lateral to ICA above and to CCA below.
- **End:** It joins with the **subclavian vein** to form **brachiocephalic vein** behind the sternoclavicular joint.
- **Tributaries:** Inferior petrosal sinus, Common facial vein, Lingual vein, Pharyngeal veins, Superior & middle thyroid veins.

4- Subclavian vein:

- Subclavian vein is a direct continuation of the **axillary vein** at the outer border of the 1st rib.
- It joins with the **internal jugular vein** to form **brachiocephalic vein** behind the sternoclavicular joint.
- The only Tributary is the **external jugular vein**.



Lymph nodes of the head:

1- Occipital lymph nodes:

Collect lymph from the occipital area of the scalp.

2- Mastoid lymph nodes: Collect lymph from the posterior neck and back of external auditory meatus.

3- Pre-auricular: They are located anterior to the auricle of the ear. Collect lymph from auricle and scalp above auricle.

4- Buccal lymph nodes:

Along with the anterior facial vein and drain cheeks.

5- Submental lymph nodes:

Collect lymph from the central lower lip, floor of mouth & tip of tongue.

6- Submandibular lymph nodes: in digastric triangle.

Collect lymph from the cheeks, the lateral side of nose, upper lip, lateral parts of lower lip and anterior tongue.

7- Retropharyngeal nodes: Collect lymph from pharynx, auditory tube and sphenoid air sinus. **N.B.** All previous groups drain into **deep lateral cervical lymph nodes**.

Lymph nodes of the neck

1- Superficial anterior cervical lymph nodes:

- Located along the anterior jugular vein.
- Receive lymph from the skin of infrahyoid region.
- Drain into the deep lateral cervical lymph nodes.

2- deep anterior cervical lymph nodes:

- These are **pre-laryngeal, paratracheal** and **pre-tracheal lymph nodes**.
- Receive from the deep structures of the neck.
- Drain into the deep lateral cervical lymph nodes.

3- Superficial lateral cervical lymph nodes:

- Located along the external jugular vein.
- Receive lymph from the parotid nodes.
- Drain into supraclavicular nodes at root of the neck.

4- Deep lateral cervical lymph nodes

- Located along the internal jugular vein.
- They are divided into **upper and lower groups**.
- The upper deep lateral cervical lymph nodes are located above superior belly of omohyoid muscle (**jugulo-digastric lymph nodes**).
- The lower deep lateral cervical nodes are located below superior belly of omohyoid muscle (**jugulo-omohyoid lymph nodes**).
- **Efferent vessels** form **jugular trunks** which **drain** into the **right lymphatic** and **thoracic ducts**.